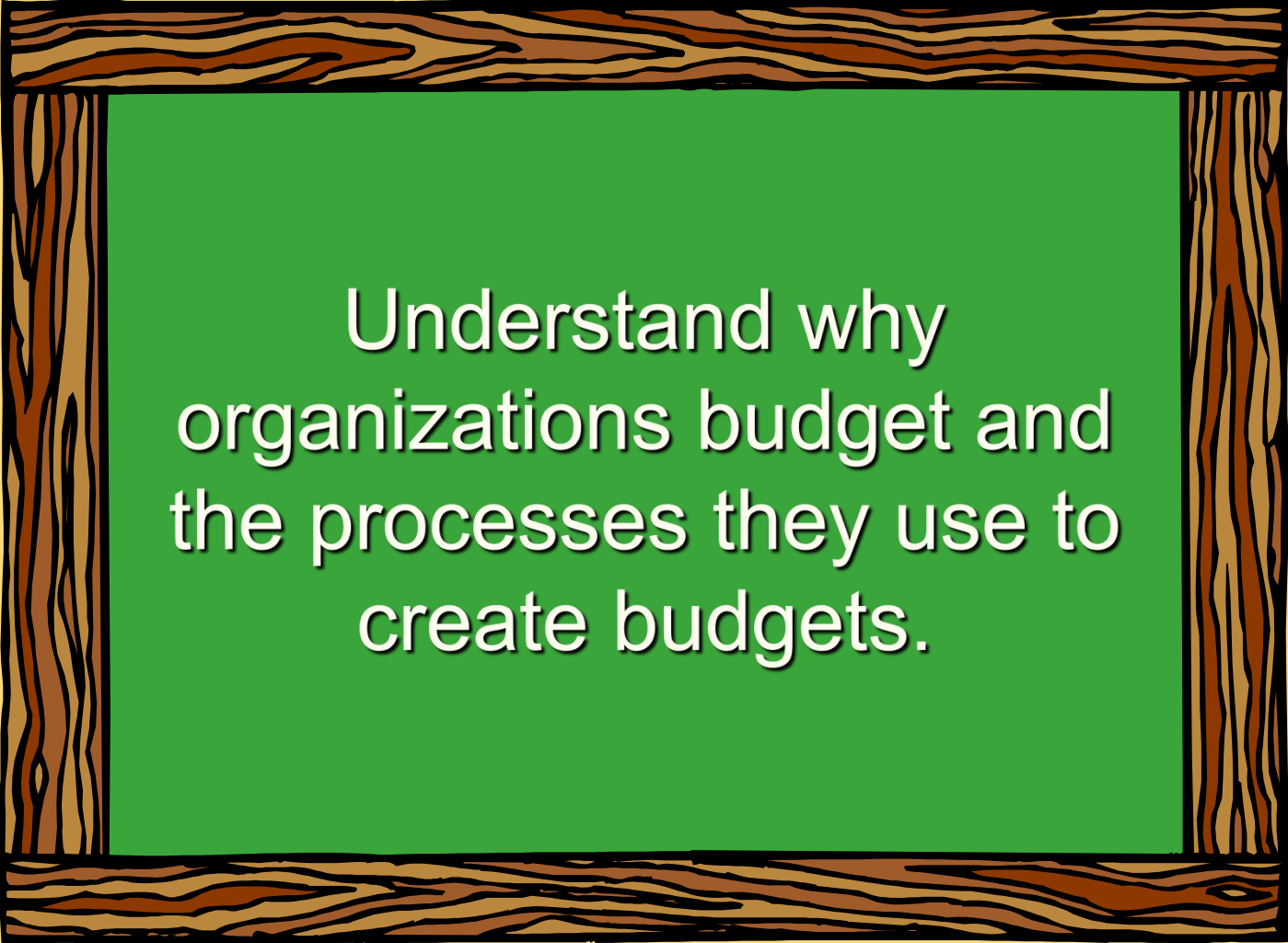


Profit Planning

Chapter Nine



Learning Objective 1

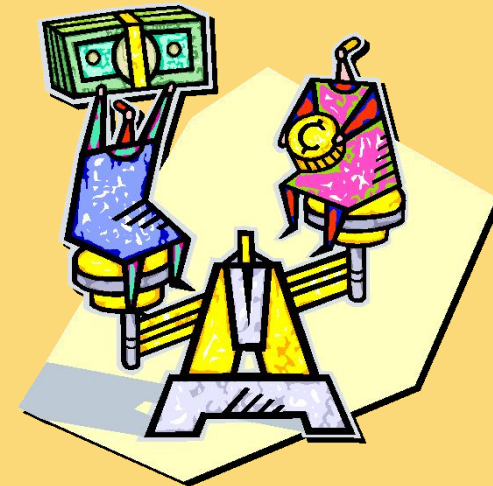


Understand why
organizations budget and
the processes they use to
create budgets.

The Basic Framework of Budgeting

A **budget** is a detailed quantitative plan for acquiring and using financial and other resources over a specified forthcoming time period.

1. The act of preparing a budget is called **budgeting**.
2. The use of budgets to control an organization's activity is known as **budgetary control**.



Planning and Control

Planning –

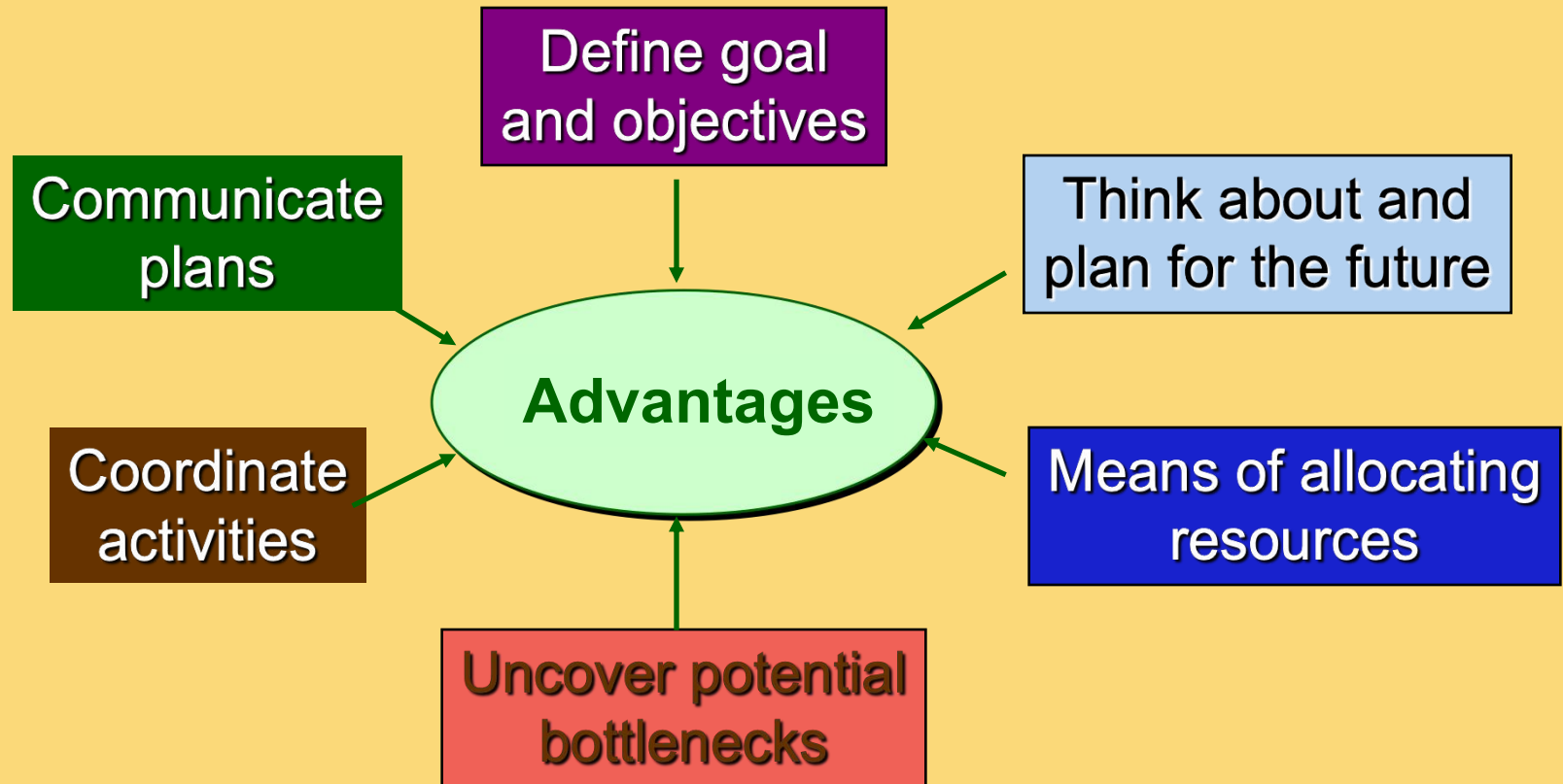
involves developing objectives and preparing various budgets to achieve these objectives.

Control –

involves the steps taken by management that attempt to ensure the objectives are attained.



Advantages of Budgeting



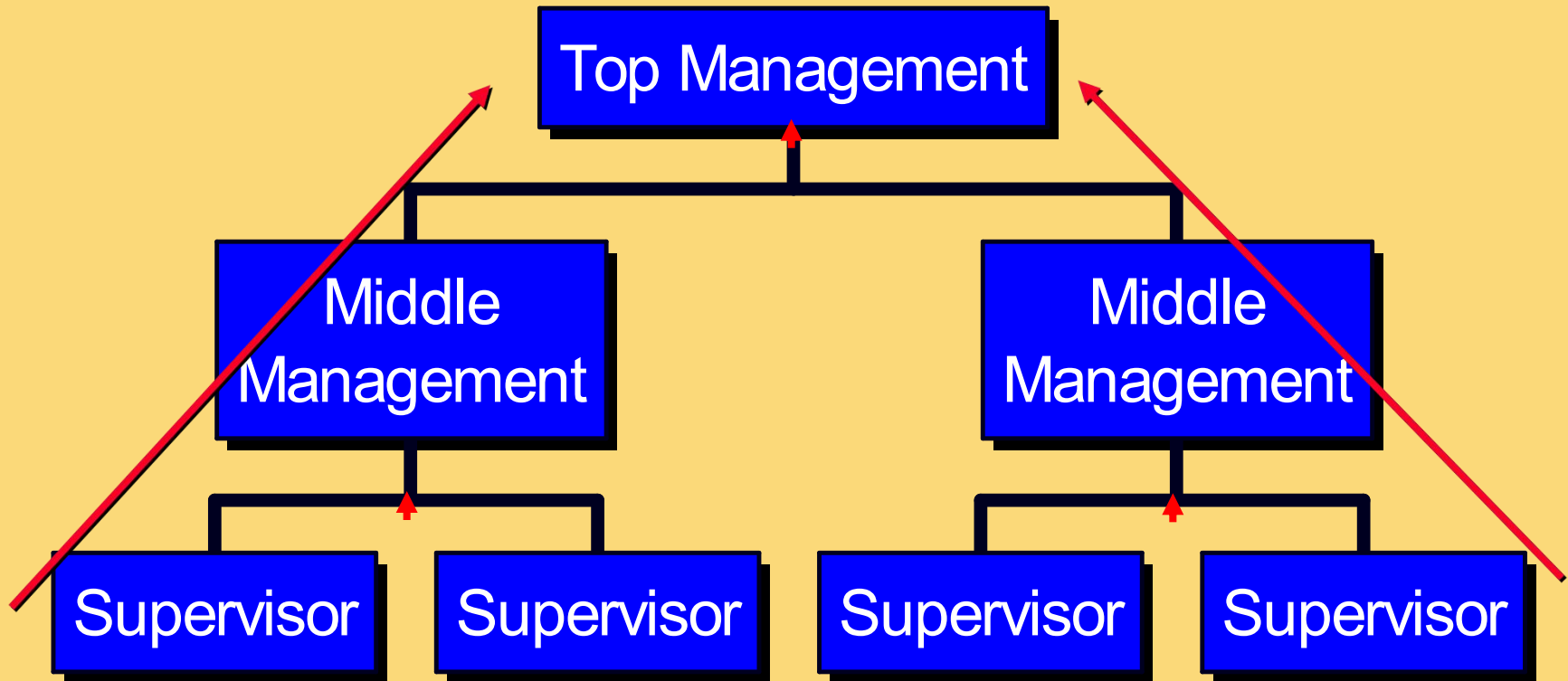
Choosing the Budget Period



The annual operating budget may be divided into quarterly or monthly budgets.

A continuous budget is a 12-month budget that rolls forward one month (or quarter) as the current month (or quarter) is completed.

Self-Imposed Budget



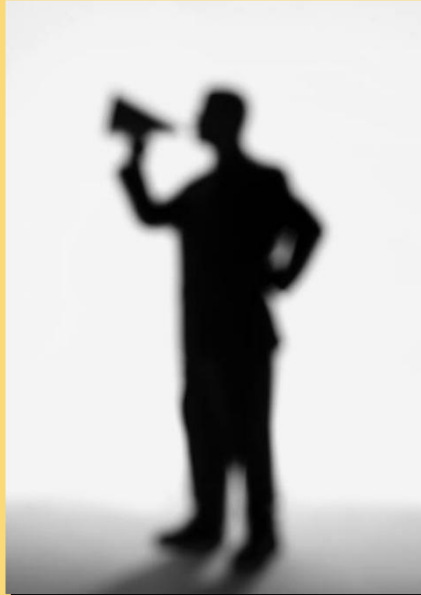
A budget is prepared with the full cooperation and participation of managers at all levels. A participative budget is also known as a **self-imposed budget**.

Advantages of Self-Imposed Budgets

1. Individuals at all levels of the organization are viewed as members of the team whose judgments are valued by top management.
2. Budget estimates prepared by front-line managers are often more accurate than estimates prepared by top managers.
3. Motivation is generally higher when individuals participate in setting their own goals than when the goals are imposed from above.
4. A manager who is not able to meet a budget imposed from above can claim that it was unrealistic. Self-imposed budgets eliminate this excuse.

Self-Imposed Budgets

Most companies do not rely exclusively upon self-imposed budgets in the sense that top managers usually initiate the budget process by issuing broad guidelines in terms of overall profits or sales.



Human Factors in Budgeting

The success of budgeting depends upon three important factors:

1. Top management must be enthusiastic and committed to the budget process.
2. Top management must not use the budget to pressure employees or blame them when something goes wrong.
3. Highly achievable budget targets are usually preferred when managers are rewarded based on meeting budget targets.

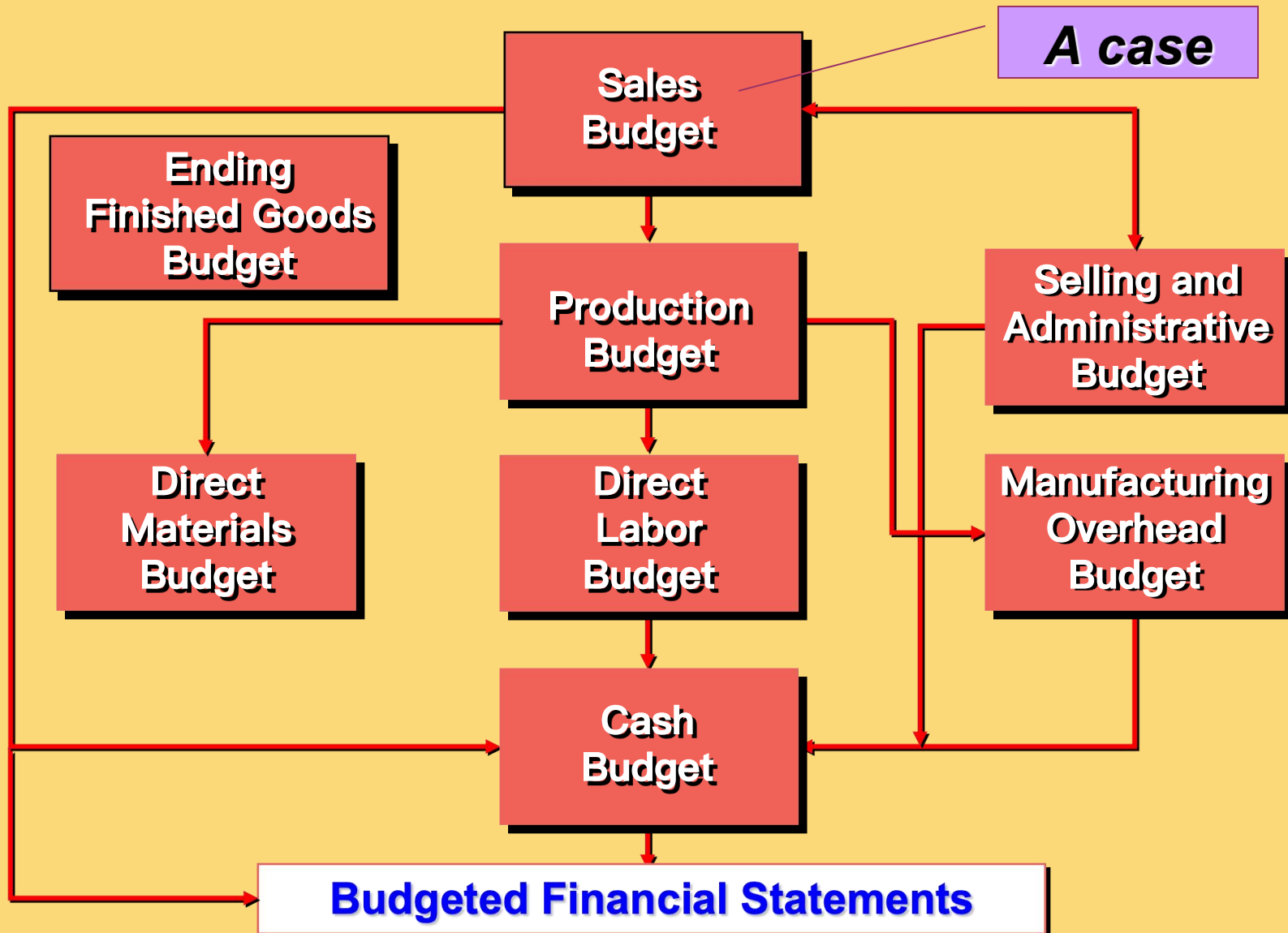
The Budget Committee

A standing committee responsible for

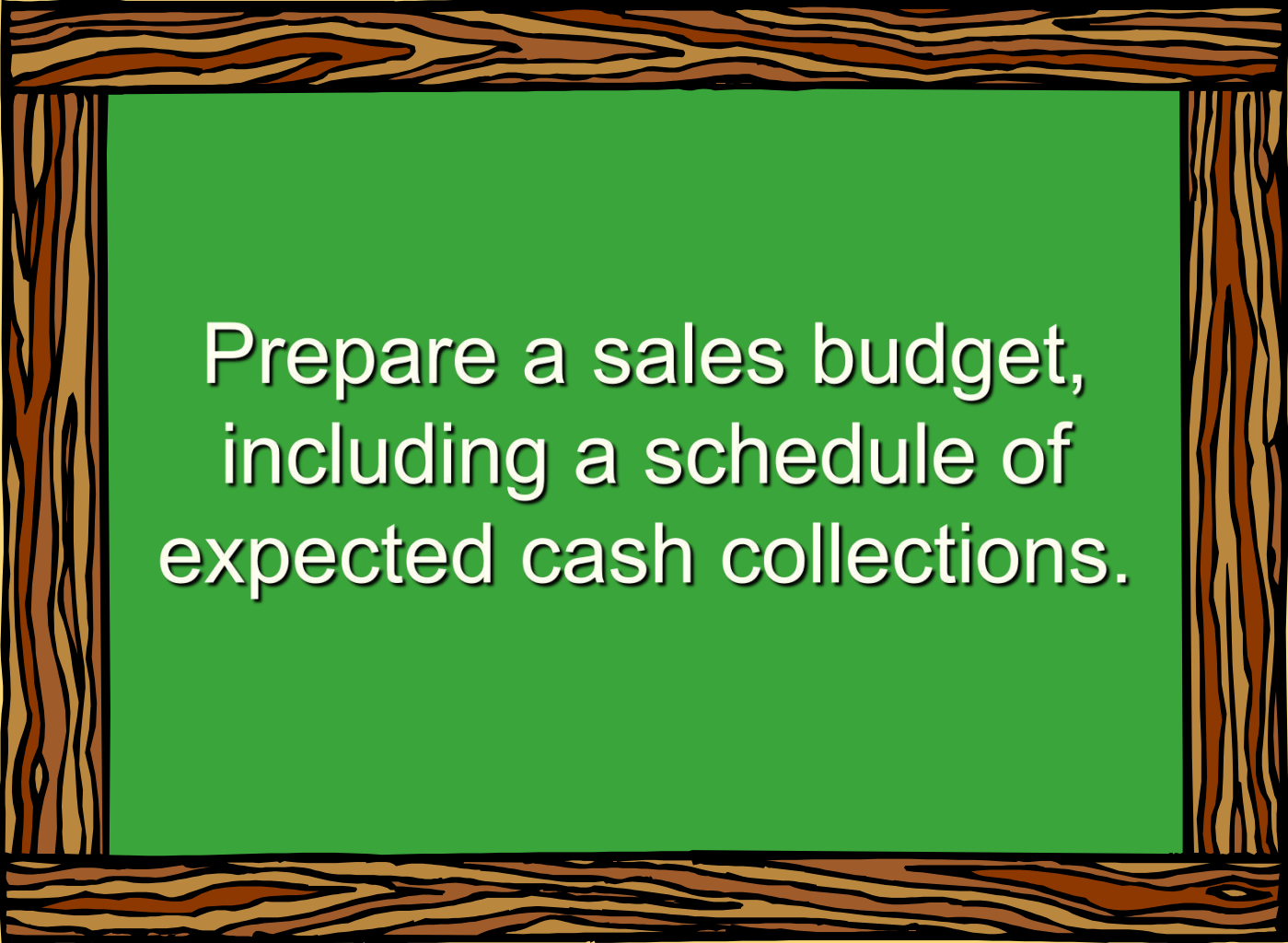
- ◆ overall policy matters relating to the budget
- ◆ coordinating the preparation of the budget



The Master Budget: An Overview



Learning Objective 2



Prepare a sales budget,
including a schedule of
expected cash collections.

Budgeting Example

① Royal Company is preparing budgets for the quarter ending June 30.

② Budgeted sales for the next five months are:

April **20,000 units**

May **50,000 units**

June **30,000 units**

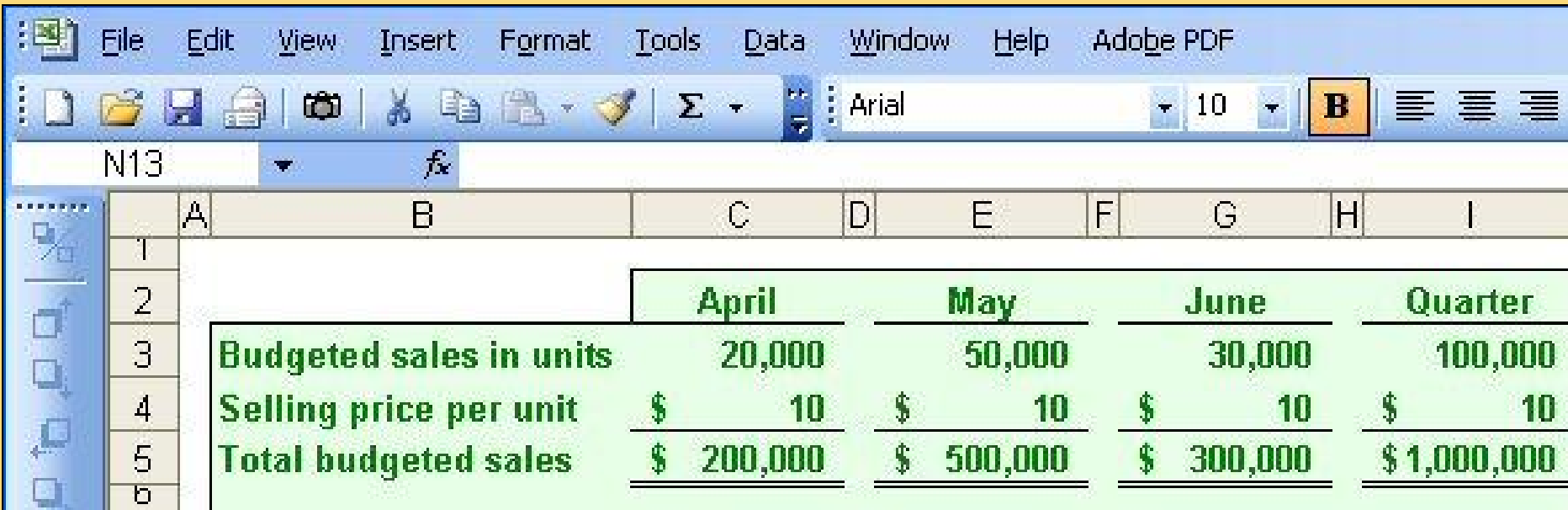
July **25,000 units**

August **15,000 units.**

③ The selling price is \$10 per unit.

The Sales Budget

The individual months of April, May, and June are summed to obtain the total projected sales in units and dollars for the quarter ended June 30th



The screenshot shows a spreadsheet application with a menu bar (File, Edit, View, Insert, Format, Tools, Data, Window, Help, Adobe PDF) and a toolbar. The active cell is N13. The spreadsheet contains a table with the following data:

	A	B	C	D	E	F	G	H	I
1									
2			April		May		June		Quarter
3		Budgeted sales in units	20,000		50,000		30,000		100,000
4		Selling price per unit	\$ 10		\$ 10		\$ 10		\$ 10
5		Total budgeted sales	\$ 200,000		\$ 500,000		\$ 300,000		\$ 1,000,000
6									

Expected Cash Collections

- All sales are on account.
- Royal's collection pattern is:
 - 70% collected in the month of sale,
 - 25% collected in the month following sale,
 - 5% uncollectible.
- The March 31 accounts receivable balance of \$30,000 will be collected in full.

Expected Cash Collections

M25										
	A	B	C	D	E	F	G	H	I	J
1										
8			April	May	June	Quarter				
9		Accounts receivable 3/31	\$ 30,000					\$ 30,000		
10										
11										
12										
13										
14										
15										
16										
17										
18		Total cash collections								
19										

Expected Cash Collections

N26										
	A	B	C	D	E	F	G	H	I	J
1										
8			April		May		June		Quarter	
9		Accounts receivable 3/31	\$ 30,000						\$ 30,000	
10		April Sales								
11		70% x \$200,000	140,000						140,000	
12		25% x \$200,000			50,000				50,000	
13										
14										
15										
16										
17										
18		Total cash collections	\$ 170,000							
19										

From the Sales Budget for April.

Expected Cash Collections

File Edit View Insert Format Tools Data Window Help Adobe PDF

N23 fx Arial 10 B

	A	B	C	D	E	F	G	H	I	J
7										
8			April		May		June		Quarter	
9		Accounts receivable 3/31	\$ 30,000						\$ 30,000	
10		April Sales								
11		70% x \$200,000	140,000						140,000	
12		25% x \$200,000			50,000				50,000	
13		May Sales								
14		70% x \$500,000			350,000				350,000	
15		25% x \$500,000					125,000		125,000	
16										
17										
18		Total cash collections	\$ 170,000	=	\$ 400,000	=				
19										

From the Sales Budget for May.

Quick Check ✓

What will be the total cash collections for the quarter?

- a. \$700,000
- b. \$220,000
- c. \$190,000
- d. \$905,000



Quick Check ✓

What will be the total cash collections for the quarter?

a. \$700,000

b. \$220,000

c. \$190,000

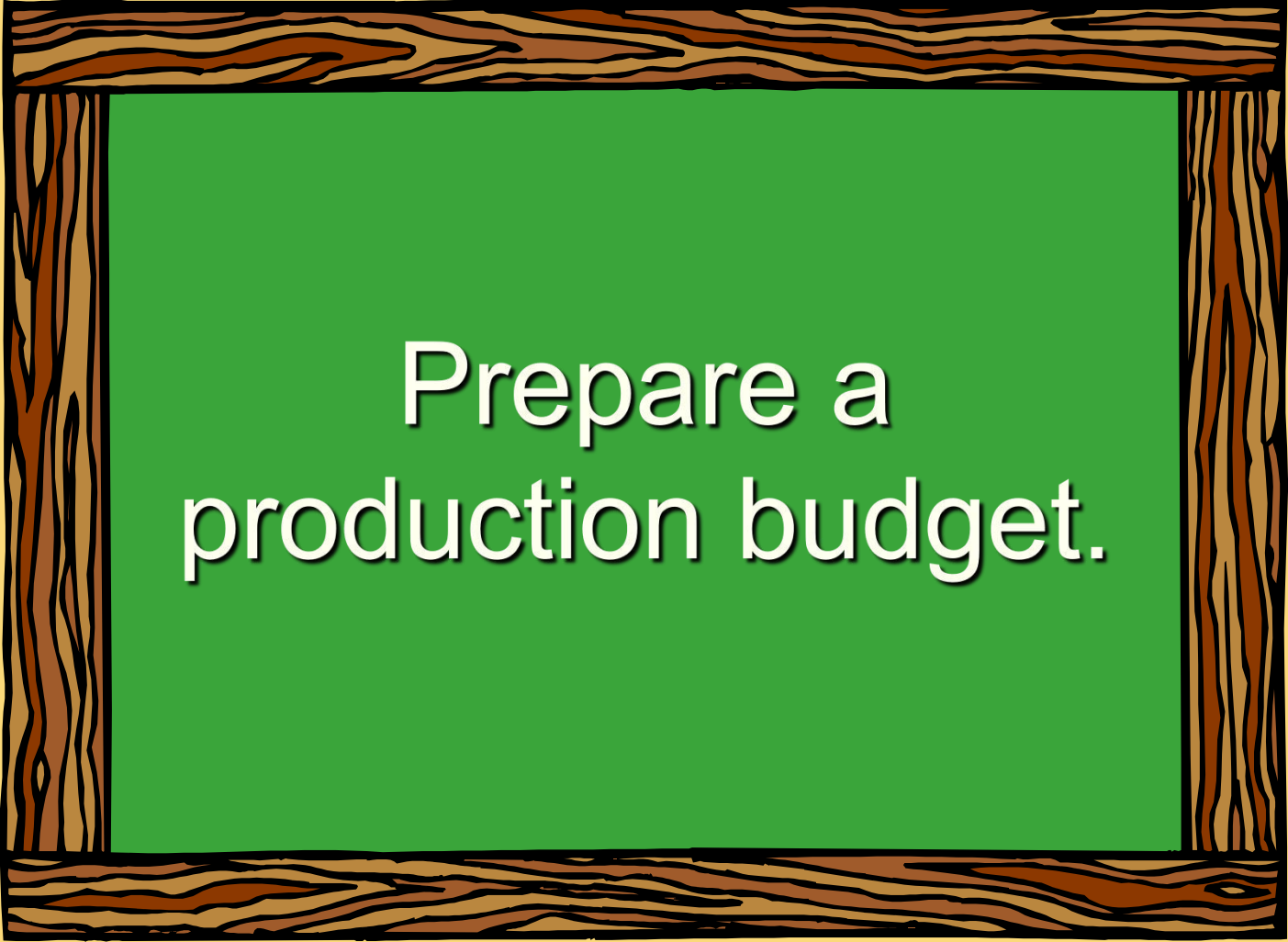
d. \$905,000



Expected Cash Collections

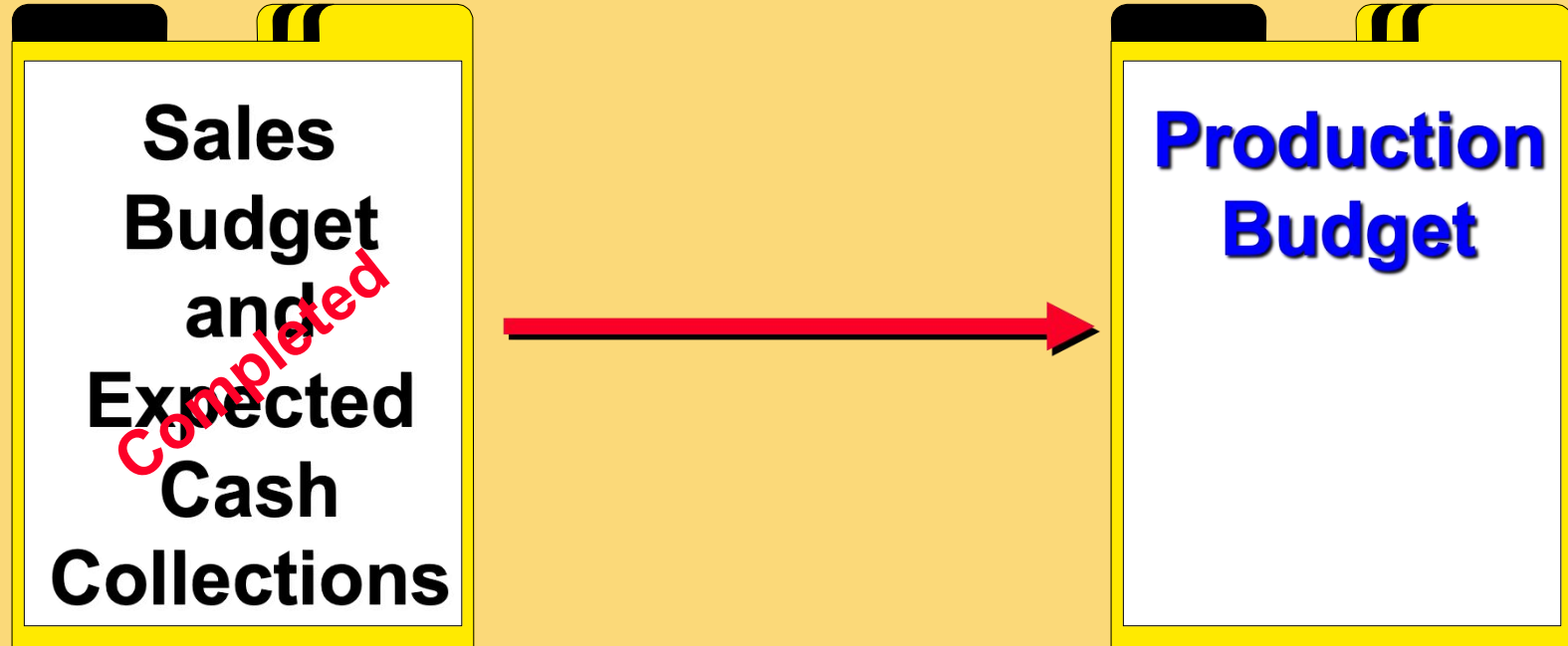
File Edit View Insert Format Tools Data Window Help Adobe PDF										
Arial 10 B										
M29 fx										
	A	B	C	D	E	F	G	H	I	J
7										
8			April		May		June		Quarter	
9		Accounts receivable 3/31	\$ 30,000						\$ 30,000	
10		April Sales								
11		70% x \$200,000	140,000						140,000	
12		25% x \$200,000			50,000				50,000	
13		May Sales								
14		70% x \$500,000			350,000				350,000	
15		25% x \$500,000					125,000		125,000	
16		Jun Sales								
17		70% x \$300,000					210,000		210,000	
18		Total cash collections	\$ 170,000		\$ 400,000		\$ 335,000		\$ 905,000	
19										

Learning Objective 3



Prepare a
production budget.

The Production Budget



Production must be adequate to meet budgeted sales and provide for sufficient ending inventory.

The Production Budget

- The management at Royal Company wants ending inventory to be equal to 20% of the following month's budgeted sales in units.
- On March 31, 4,000 units were on hand.

Let's prepare the production budget.

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The Production Budget

File Edit View Insert Format Tools Data Window Help Adobe PDF										
Arial 10 B										
N43 fx										
	A	B	C	D	E	F	G	H	I	J
1										
21			April	May	June	Quarter				
22		Budgeted Sales	20,000	50,000	30,000	100,000				
23		Add: Desired ending inventory	10,000							
24		Total Needs	30,000							
25		Less: Beginning inventory	4,000							
26		Required production	26,000							
27										

**March 31
ending inventory**

Budgeted May sales	50,000
Desired ending inventory %	20%
Desired ending inventory	<u>10,000</u>

Quick Check ✓

What is the required production for May?

- a. 56,000 units
- b. 46,000 units
- c. 62,000 units
- d. 52,000 units



Quick Check ✓

What is the required production for May?

a. 56,000 units

b. 46,000 units

c. 62,000 units

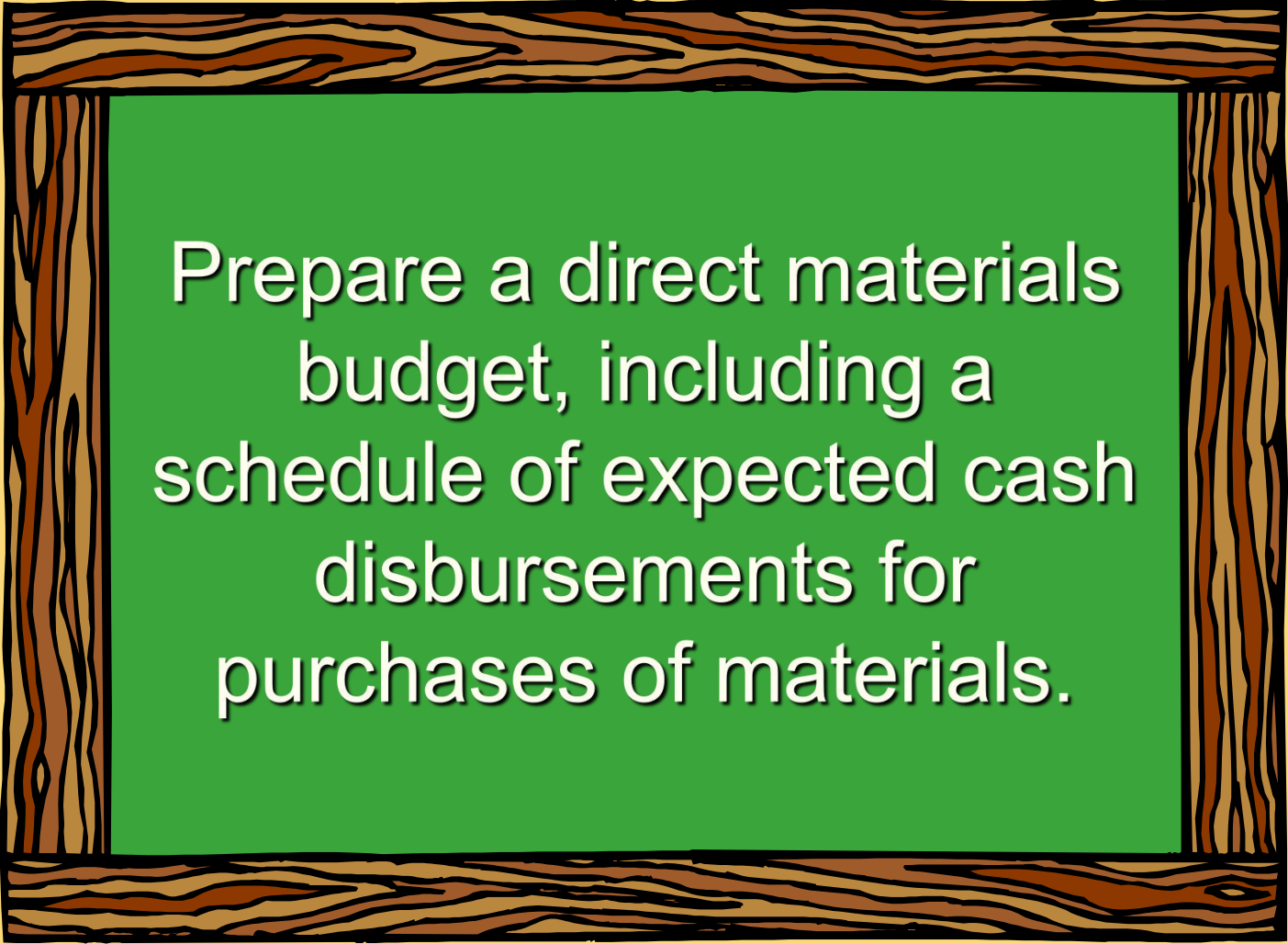
d. 52,000 units



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Assumed ending inventory.

Learning Objective 4



Prepare a direct materials budget, including a schedule of expected cash disbursements for purchases of materials.

The Direct Materials Budget

- At Royal Company, *five pounds* of material are required per unit of product.
 - Management wants materials on hand at the end of each month equal to *10%* of the following month's production.
 - On March 31, 13,000 pounds of material are on hand. Material cost is *\$0.40* per pound.
- Let's prepare the direct materials budget.

The Direct Materials Budget

M48										
	A	B	C	D	E	F	G	H	I	J
28			April	May	June	Quarter				
29		Required production	26,000	46,000	29,000	101,000				
30		Materials per unit (pounds)								
31		Production needs								
32		Add: Desired ending inventory								
33		Total needed								
34		Less: Beginning inventory								
35		Materials to be purchased								
36										

From production budget

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The Direct Materials Budget

N50										
	A	B	C	D	E	F	G	H	I	J
28			April		May		June		Quarter	
29		Production	26,000		46,000		29,000		101,000	
30		Materials per unit (pounds)	5		5		5		5	
31		Production needs	130,000		230,000		145,000		505,000	
32		Add: Desired ending inventory	23,000							
33		Total needed	153,000							
34		Less: Beginning inventory	13,000							
35		Materials to be purchased	140,000		?					
36										

10% of following month's production needs.

Calculate the materials to be purchased in May.

Quick Check ✓

How much materials should be purchased in May?

- a. 221,500 pounds
- b. 240,000 pounds
- c. 230,000 pounds
- d. 211,500 pounds



Quick Check ✓

How much materials should be purchased in May?

- ☒ a. 221,500 pounds
- ☐ b. 240,000 pounds
- ☐ c. 230,000 pounds
- ☐ d. 211,500 pounds



The Direct Materials Budget

N161										
	A	B	C	D	E	F	G	H	I	J
27										
28			April		May		June		Quarter	
29	Production		26,000		46,000		29,000		101,000	
30	Materials per unit (pounds)		5		5		5		5	
31	Production needs		130,000		230,000		145,000		505,000	
32	Add: Desired ending invento		23,000		14,500					
33	Total needed		153,000		244,500					
34	Less: Beginning inventory		13,000		23,000					
35	Materials to be purchased		140,000		221,500					
36										

The Direct Materials Budget

M51									
	A	B	C	D	E	F	G	H	I
28			April		May		June		Quarter
29		Production	26,000		46,000		29,000		101,000
30		Materials per unit (pounds)	5		5		5		5
31		Production needs	130,000		230,000		145,000		505,000
32		Add: Desired ending inventory	23,000		14,500		11,500		11,500
33		Total needed	153,000		244,500		156,500		516,500
34		Less: Beginning inventory	13,000		23,000		14,500		13,000
35		Materials to be purchased	140,000		221,500		142,000		503,500
36									

Assumed ending inventory

Expected Cash Disbursement for Materials

- Royal pays *\$0.40 per pound* for its materials.
- *One-half* of a month's purchases is paid for in the month of purchase; the other half is paid in the following month.
- The March 31 accounts payable balance is \$12,000.

Let's calculate expected cash disbursements.

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Expected Cash Disbursement for Materials

M61										
	A	B	C	D	E	F	G	H	I	J
1										
28			April		May		June		Quarter	
37		Accounts payable 3/31	\$ 12,000						\$ 12,000	
38		April purchases								
39		50% x \$56,000	28,000						28,000	
40		50% x \$56,000			28,000				28,000	
41										
42										
43										
44										
45										
46		Total cash disbursements								
47										

Compute the expected cash disbursements for materials for the quarter.

$$140,000 \text{ lbs.} \times \$.40/\text{lb.} = \$56,000$$

Quick Check ✓

What are the total cash disbursements for the quarter?

- a. \$185,000
- b. \$ 68,000
- c. \$ 56,000
- d. \$201,400



Quick Check ✓

What are the total cash disbursements for the quarter?

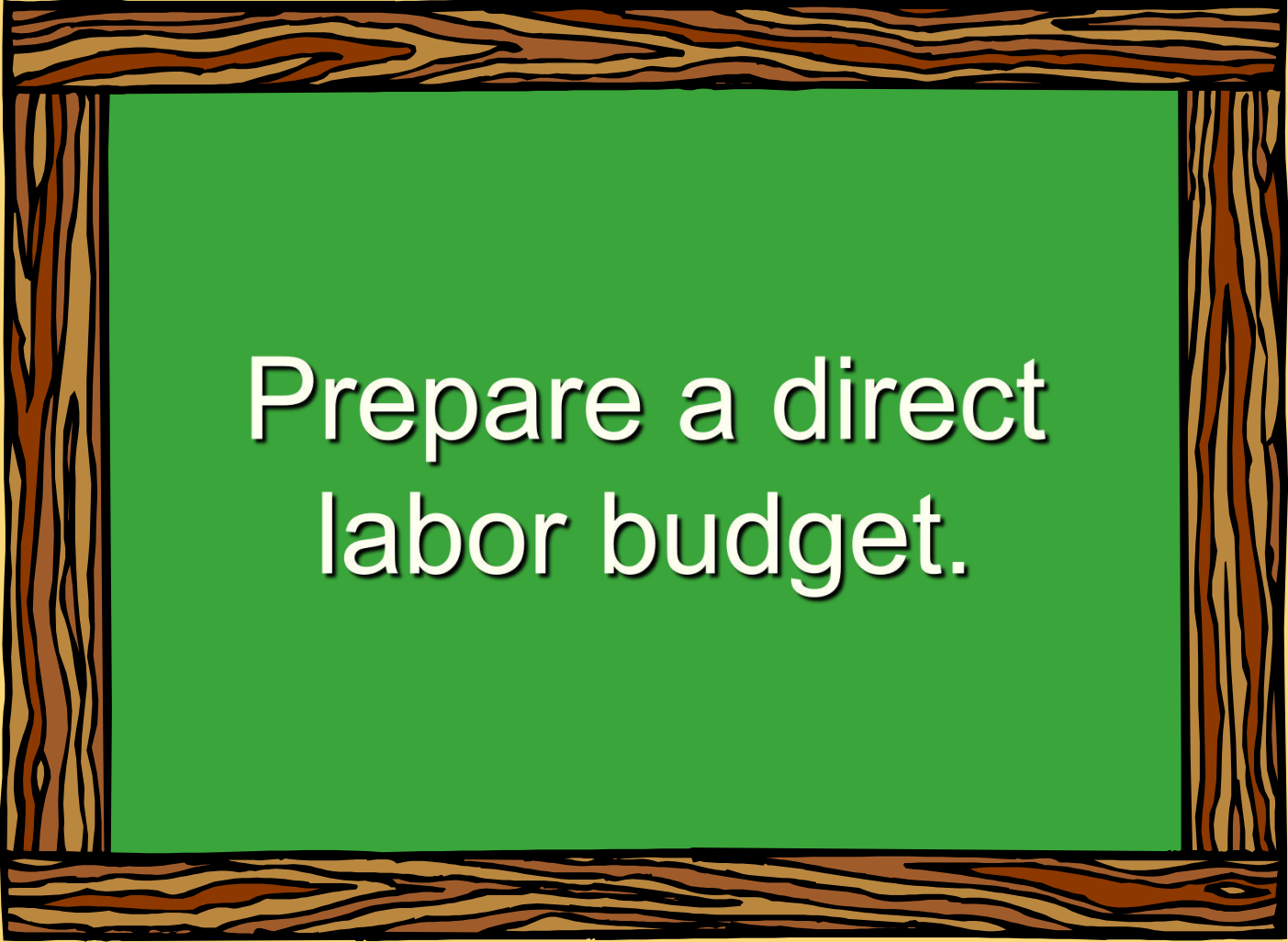
- a. \$185,000
- b. \$ 68,000
- c. \$ 56,000
- d. \$201,400



Expected Cash Disbursement for Materials

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Arial 10 B										
L61 fx										
	A	B	C	D	E	F	G	H	I	J
1										
28			April		May		June		Quarter	
37		Accounts payable 3/31	\$ 12,000						\$ 12,000	
38		April purchases								
39		50% x \$56,000	28,000						28,000	
40		50% x \$56,000			28,000				28,000	
41		May purchases								
42		50% x \$88,600			44,300				44,300	
43		50% x \$88,600					44,300		44,300	
44		June purchases								
45		50% x \$56,800					28,400		28,400	
46		Total cash disbursements	\$ 40,000		\$ 72,300		\$ 72,700		\$ 185,000	
47										

Learning Objective 5



Prepare a direct
labor budget.

The Direct Labor Budget

- At Royal, each unit of product requires 0.05 hours (3 minutes) of direct labor.
- The Company has a “no layoff” policy so all employees will be paid for 40 hours of work each week at least.
- In exchange for the “no layoff” policy, workers agree to a wage rate of \$10 per hour regardless of the hours worked (no overtime pay).
- For the next three months, the direct labor workforce will be paid for a minimum of 1,500 hours per month.

Let's prepare the direct labor budget.

The Direct Labor Budget

File Edit View Insert Format Tools Data Window Help Adobe PDF										
Arial 10 B										
L70	fx									
	A	B	C	D	E	F	G	H	I	J
1										
28			April	May	June		Quarter			
48		Units of production	26,000	46,000	29,000		101,000			
49		Direct labor per unit								
50		Labor hours required								
51										
52										
53										
54		Total direct labor costs								
55										

From production budget.

The Direct Labor Budget

File Edit View Insert Format Tools Data Window Help Adobe PDF Arial 10 B										
L70	fx									
	A	B	C	D	E	F	G	H	I	J
1										
28			April		May		June		Quarter	
48		Units of production	26,000		46,000		29,000		101,000	
49		Direct labor per unit	0.05		0.05		0.05		0.05	
50		Labor hours required	1,300		2,300		1,450		5,050	
51										
52										
53										
54		Total direct labor costs								
55										

The Direct Labor Budget

File Edit View Insert Format Tools Data Window Help Adobe PDF										
Arial 10 B										
N70										
	A	B	C	D	E	F	G	H	I	J
28			April	May	June		Quarter			
48		Units of production	26,000	46,000	29,000		101,000			
49		Direct labor per unit	0.05	0.05	0.05		0.05			
50		Labor hours required	1,300	2,300	1,450		5,050			
51		Guaranteed labor hours	1,500	1,500	1,500					
52		Labor hours paid	1,500	2,300	1,500		5,300			
53										
54		Total direct labor costs								
55										

Greater of labor hours required
or labor hours guaranteed.

The Direct Labor Budget

File Edit View Insert Format Tools Data Window Help M69 fx Arial 10 B										
	A	B	C	D	E	F	G	H	I	J
28			April		May		June		Quarter	
48		Units of production	26,000		46,000		29,000		101,000	
49		Direct labor per unit	0.05		0.05		0.05		0.05	
50		Labor hours required	1,300		2,300		1,450		5,050	
51		Guaranteed labor hours	1,500		1,500		1,500			
52		Labor hours paid	1,500		2,300		1,500		5,300	
53		Hourly wage rate	\$ 10		\$ 10		\$ 10		\$ 10	
54		Total direct labor costs	\$ 15,000		\$ 23,000		\$ 15,000		\$ 53,000	
55										

Quick Check ✓

What would be the total direct labor cost for the quarter if the company follows its no lay-off policy, but pays \$15 (time-and-a-half) for every hour worked in excess of 1,500 hours in a month?

- a. \$79,500
- b. \$64,500
- c. \$61,000
- d. \$57,000



Quick Check ✓

What would be the total direct labor cost for the quarter if the company follows its no lay-off policy, but pays \$15 (time-and-a-half) for every hour worked in excess of regular hours?

a. \$79,500

b. \$64,500

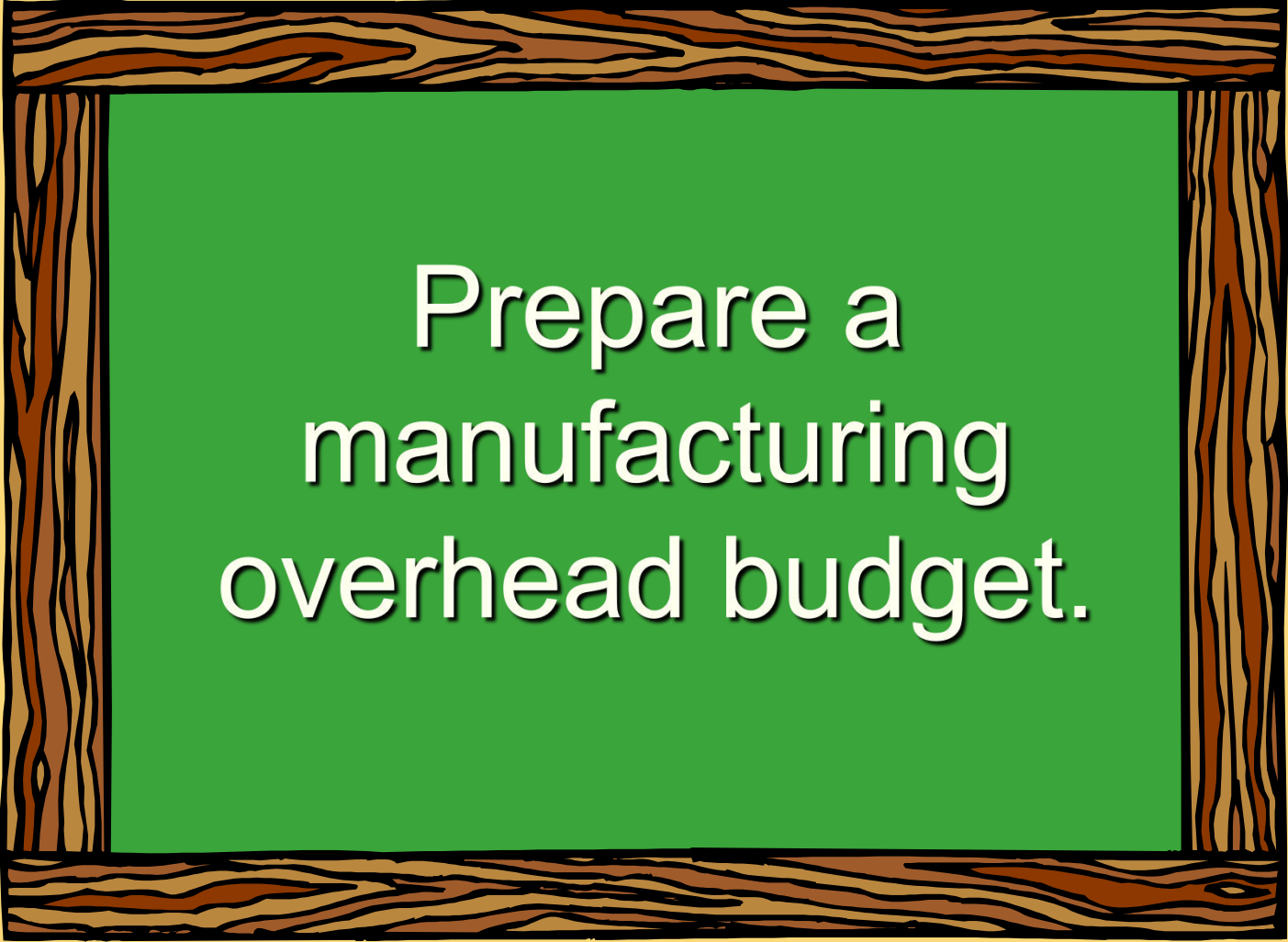
c. \$61,000

d. \$57,000

	<i>April</i>	<i>May</i>	<i>June</i>	<i>Quarter</i>
Labor hours required	1,300	2,300	1,450	
Regular hours paid	<u>1,500</u>	<u>1,500</u>	<u>1,500</u>	4,500
Overtime hours paid	-	800	-	800
Total regular hours	4,500	\$10	\$45,000	
Total overtime hours	800	\$15	<u>\$12,000</u>	
Total pay			<u>\$57,000</u>	



Learning Objective 6



Prepare a
manufacturing
overhead budget.

Manufacturing Overhead Budget

- At Royal, manufacturing overhead is applied to units of product on the basis of direct labor hours.
- The variable manufacturing overhead rate is \$20 per direct labor hour.
- Fixed manufacturing overhead is \$50,000 per month and includes \$20,000 of noncash costs (primarily depreciation of plant assets).

Let's prepare the manufacturing overhead budget.

Manufacturing Overhead Budget

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O108		fx								
	A	B	C	D	E	F	G	H	I	J
84										
85										
86										
87										
88										
89										
90										
91										
92										
93										
94										

	April	May	June	Quarter
Budgeted DLH	1,300	2,300	1,450	5,050
Variable mfg. OH rate	\$ 20	\$ 20	\$ 20	\$ 20
Variable mfg. OH costs	\$ 26,000	\$ 46,000	\$ 29,000	\$ 101,000
Fixed mfg. OH costs				
Total mfg. OH costs				
Less: noncash costs				
Cash disbursements				
for manufacturing OH				

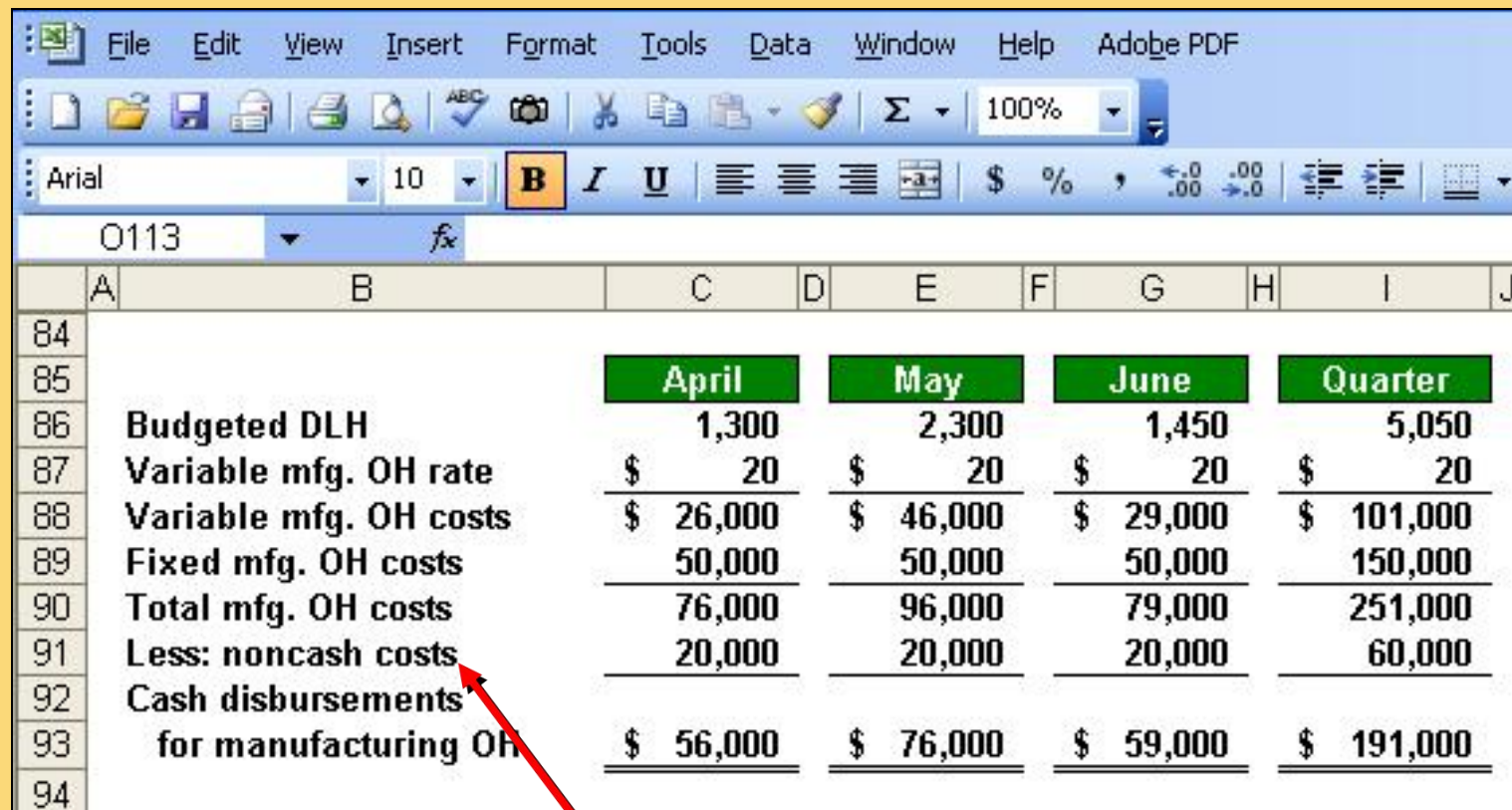
Direct Labor Budget.

Manufacturing Overhead Budget

File Edit View Insert Format Tools Data Window Help Adobe PDF										
Arial 10 B I U										
N103 fx										
	A	B	C	D	E	F	G	H	I	J
84										
85			April	May	June	Quarter				
86		Budgeted DLH	1,300	2,300	1,450	5,050				
87		Variable mfg. OH rate	\$ 20	\$ 20	\$ 20	\$ 20				
88		Variable mfg. OH costs	\$ 26,000	\$ 46,000	\$ 29,000	\$ 101,000				
89		Fixed mfg. OH costs	50,000	50,000	50,000	150,000				
90		Total mfg. OH costs	76,000	96,000	79,000	251,000				
91		Less: non	$\frac{\text{Total mfg. OH for quarter } \$251,000}{\text{Total labor hours required } 5,050} = \$49.70 \text{ per hour } *$							
92		Cash disb								
93		for man								
94										

* rounded

Manufacturing Overhead Budget



	A	B	C	D	E	F	G	H	I	J
84										
85			April	May	June	Quarter				
86		Budgeted DLH	1,300	2,300	1,450	5,050				
87		Variable mfg. OH rate	\$ 20	\$ 20	\$ 20	\$ 20				
88		Variable mfg. OH costs	\$ 26,000	\$ 46,000	\$ 29,000	\$ 101,000				
89		Fixed mfg. OH costs	50,000	50,000	50,000	150,000				
90		Total mfg. OH costs	76,000	96,000	79,000	251,000				
91		Less: noncash costs	20,000	20,000	20,000	60,000				
92		Cash disbursements								
93		for manufacturing OH	\$ 56,000	\$ 76,000	\$ 59,000	\$ 191,000				
94										

Depreciation is a noncash charge.

Ending Finished Goods Inventory Budget

<u>Production costs per unit</u>	<u>Quantity</u>	<u>Cost</u>	<u>Total</u>
Direct materials	5.00 lbs.	\$ 0.40	\$ 2.00
Direct labor			
Manufacturing overhead			
<u>Budgeted finished goods inventory</u>			
Ending inventory in units			
Unit product cost			
Ending finished goods inventory			

Direct materials
budget and information.

Ending Finished Goods Inventory Budget

<u>Production costs per unit</u>	<u>Quantity</u>	<u>Cost</u>	<u>Total</u>
Direct materials	5.00 lbs.	\$ 0.40	\$ 2.00
Direct labor	0.05 hrs.	\$10.00	0.50
Manufacturing overhead			
<u>Budgeted finished goods inventory</u>			
Ending inventory in units			
Unit product cost			
Ending finished goods inventory			

Direct labor budget.

Ending Finished Goods Inventory Budget

<u>Production costs per unit</u>	<u>Quantity</u>	<u>Cost</u>	<u>Total</u>
Direct materials	5.00 lbs.	\$ 0.40	\$ 2.00
Direct labor	0.05 hrs.	\$ 10.00	0.50
Manufacturing overhead	0.05 hrs.	\$ 49.70	2.49
			<u>\$ 4.99</u>
<u>Budgeted finished goods inventory</u>			
Ending inventory in units			
Unit product cost			\$ 4.99
Ending finished goods inventory			<u>?</u>

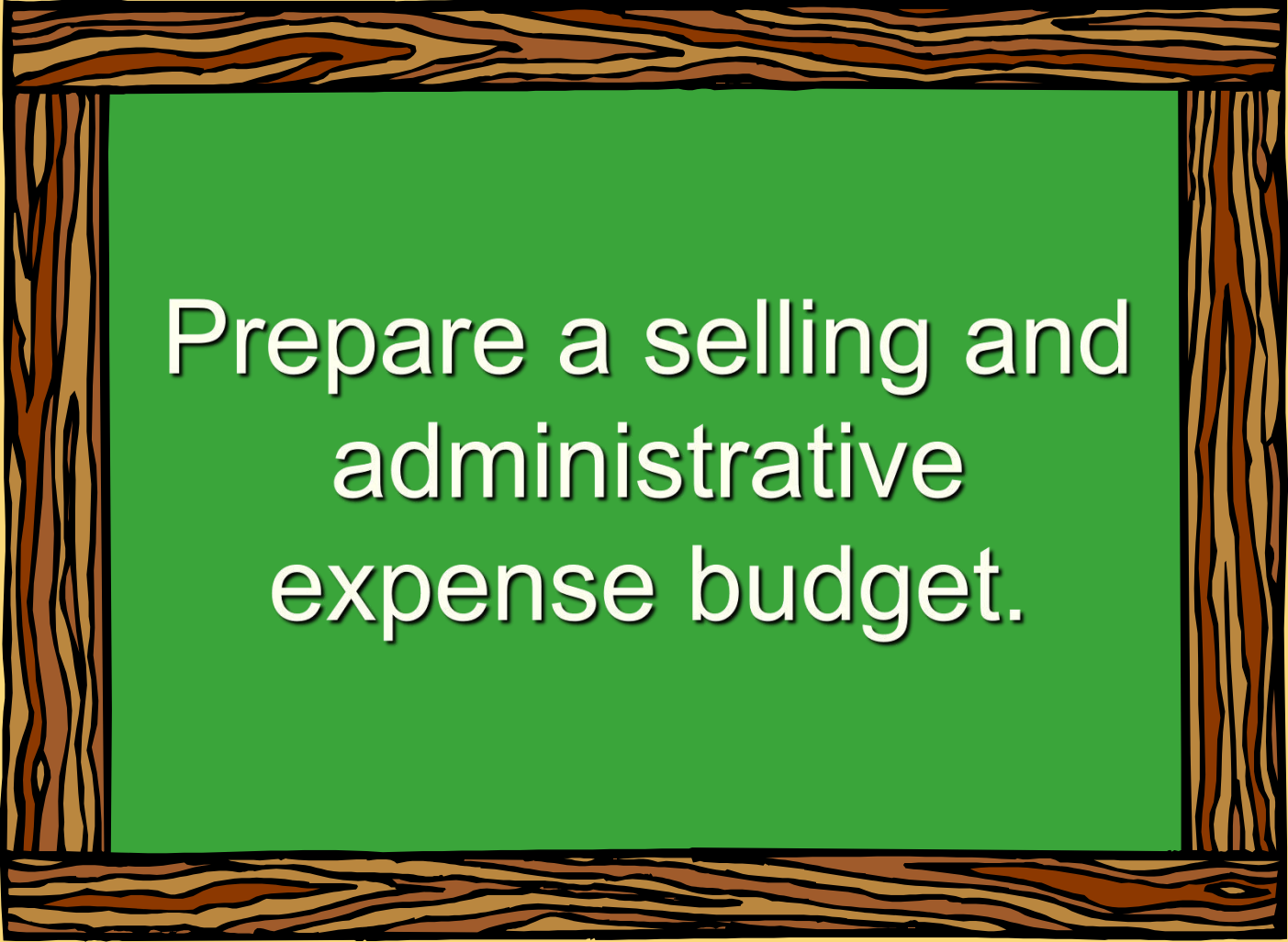
$$\frac{\text{Total mfg. OH for quarter } \$251,000}{\text{Total labor hours required } 5,050} = \$49.70 \text{ per hour } ^*$$

Ending Finished Goods Inventory Budget

<u>Production costs per unit</u>	<u>Quantity</u>	<u>Cost</u>	<u>Total</u>
Direct materials	5.00 lbs.	\$ 0.40	\$ 2.00
Direct labor	0.05 hrs.	\$10.00	0.50
Manufacturing overhead	0.05 hrs.	\$49.70	2.49
			<u>\$ 4.99</u>
<u>Budgeted finished goods inventory</u>			
Ending inventory in units			5,000
Unit product cost			\$ 4.99
Ending finished goods inventory			<u>\$24,950</u>

Production Budget.

Learning Objective 7



Prepare a selling and
administrative
expense budget.

Selling and Administrative Expense Budget

- At Royal, the selling and administrative expenses budget is divided into variable and fixed components.
- The variable selling and administrative expenses are \$0.50 per unit sold.
- Fixed selling and administrative expenses are \$70,000 per month.
- The fixed selling and administrative expenses include \$10,000 in costs – primarily depreciation – that are not cash outflows of the current month.

Let's prepare the company's selling and administrative expense budget.

Selling and Administrative Expense Budget

File Edit View Insert Format Tools Data Window Help Adobe PDF										
Arial 10 B										
NB1										
	A	B	C	D	E	F	G	H	I	J
28			April		May		June		Quarter	
48		Budgeted sales	20,000							
57		Variable S & A rate	\$ 0.50							
58		Variable expenses	\$ 10,000							
59		Fixed S & A expenses	70,000							
60		Total S & A expenses	80,000							
61		Less: Noncash expenses	10,000							
62		Cash S & A expenses	\$ 70,000						?	
63										

Calculate the selling and administrative cash expenses for the quarter.

Quick Check ✓

What are the total cash disbursements for selling and administrative expenses for the quarter?

- a. \$180,000**
- b. \$230,000**
- c. \$110,000**
- d. \$ 70,000**



Quick Check ✓

What are the total cash disbursements for selling and administrative expenses for the quarter?

a. \$180,000

☒ b. \$230,000

c. \$110,000

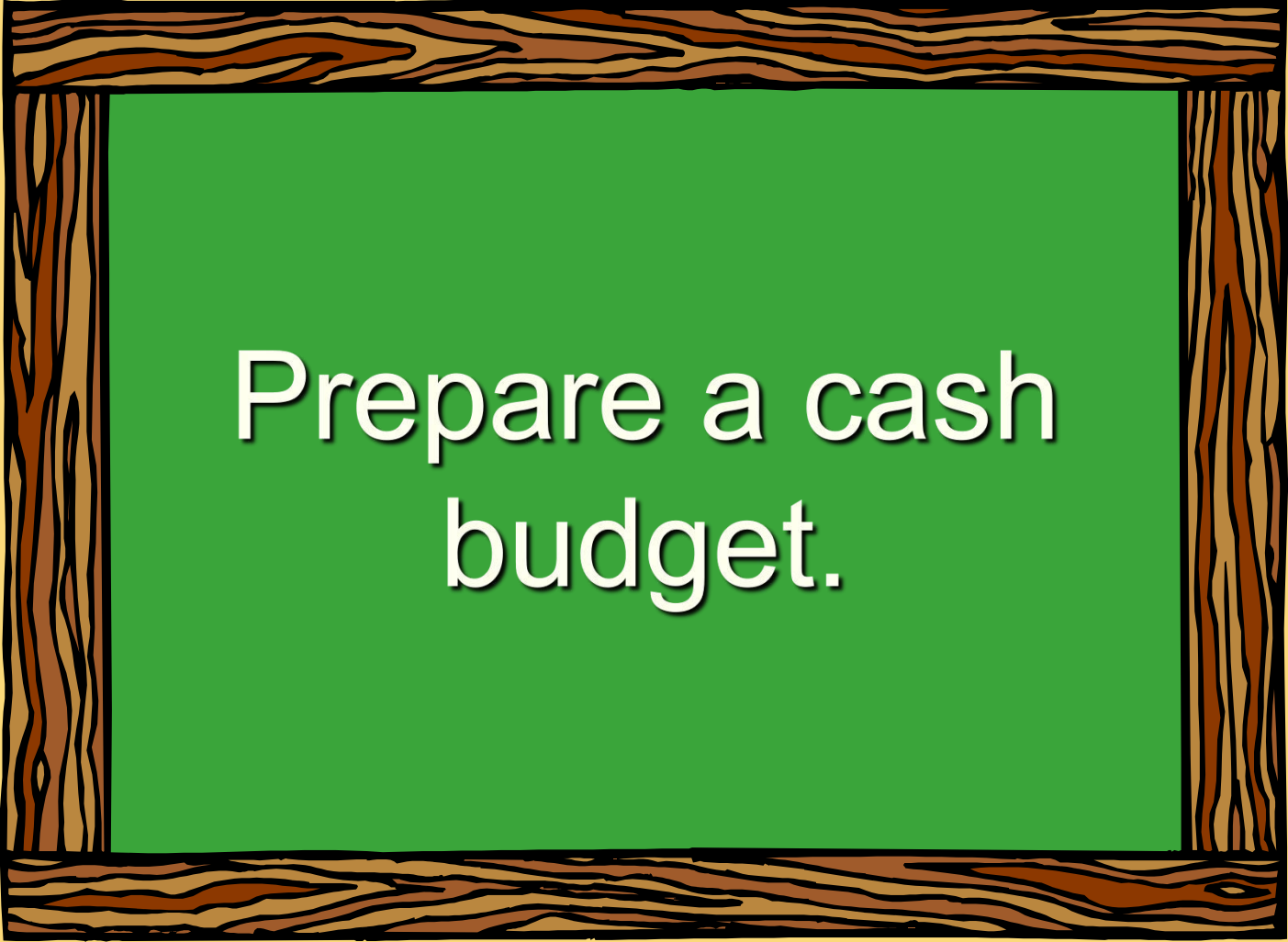
d. \$ 70,000



Selling and Administrative Expense Budget

File Edit View Insert Format Tools Data Window Help Adobe PDF										
Arial 10 B										
N78	fx									
	A	B	C	D	E	F	G	H	I	J
1										
28			April	May	June	Quarter				
48		Budgeted sales	20,000	50,000	30,000	100,000				
57		Variable S & A rate	\$ 0.50	\$ 0.50	\$ 0.50	\$ 0.50				
58		Variable expenses	\$ 10,000	\$ 25,000	\$ 15,000	\$ 50,000				
59		Fixed S & A expenses	70,000	70,000	70,000	210,000				
60		Total S & A expenses	80,000	95,000	85,000	260,000				
61		Less: Noncash expenses	10,000	10,000	10,000	30,000				
62		Cash S & A expenses	<u>\$ 70,000</u>	<u>\$ 85,000</u>	<u>\$ 75,000</u>	<u>\$ 230,000</u>				
63										

Learning Objective 8



Prepare a cash
budget.

Format of the Cash Budget

The cash budget is divided into four sections:

1. Cash receipts listing all cash inflows excluding borrowing;
2. Cash disbursements listing all payments excluding repayments of principal and interest;
3. Cash excess or deficiency; and
4. The financing section listing all borrowings, repayments and interest.

The Cash Budget

Royal:

- | Maintains a 16% open line of credit for \$75,000
- | Maintains a minimum cash balance of \$30,000
- | Borrows on the first day of the month and repays loans on the last day of the month
- | Pays a cash dividend of \$49,000 in April
- | Purchases \$143,700 of equipment in May and \$48,300 in June (both purchases paid in cash)
- | Has an April 1 cash balance of \$40,000

The Cash Budget

M90									
	A	B	C	D	E	F	G	H	I
63			April	May	June	Quarter			
64		Beginning cash balance	\$ 40,000						
65		Add: Cash collections	170,000						
66		Total cash available	210,000						
67		Less: Cash disbursements							
68		Materials							
69		Direct labor							
70		Manufacturing overhead							
71		Selling and administrative							
72		Dividend							
73		Total disbursements							
74		Excess (deficiency)							
75		Financing:							
76		Borrowing							
77		Repayments							
78		Interest							
79		Total financing							
80		Ending cash balance							
81									

Schedule of Expected Cash Collections.

The Cash Budget

M88										
	A	B	C	D	E	F	G	H	I	J
1										
63			April							
64		Beginning cash balance	\$ 40,000							
65		Add: Cash collections	170,000							
66		Total cash available	210,000							
67		Less: Cash disbursements								
68		Materials	40,000							
69		Direct labor	15,000							
70		Manufacturing overhead	56,000							
71		Selling and administrative	70,000							
72		Equipment purchase	-							
73		Dividend	49,000							
74		Total disbursements	230,000							
75		Excess (deficiency)								
76		Financing:								
77		Borrowing								
78		Repayments								
79		Interest								
80		Total financing								
81		Ending cash balance								
82										

**Schedule of Expected
Cash Disbursements.**

**Direct Labor
Budget.**

**Manufacturing
Overhead Budget.**

**Selling and Administrative
Expense Budget.**

The Cash Budget

M88									
	A	B	C	D	E	F	G	H	I
63			April	May	June	Quarter			
64		Beginning cash balance	\$ 40,000						
65		Add: Cash collections	170,000						
66		Total cash available	210,000						
67		Less: Cash disbursements							
68		Materials	40,000						
69		Direct labor	15,000						
70		Manufacturing overhead	56,000						
71		Selling and administrative	70,000						
72		Equipment purchase	-						
73		Dividend	49,000						
74		Total disbursements	230,000						
75		Excess (deficiency)	(20,000)						
76		Financing:							
77		Borrowing							
78		Repayments							
79		Interest							
80		Total financing							
81		Ending cash balance							
82									

Because Royal maintains a cash balance of \$30,000, the company must borrow \$50,000 on its line-of-credit.

The Cash Budget

M89										
	A	B	C	D	E	F	G	H	I	J
63			April	May	June	Quarter				
64		Beginning cash balance	\$ 40,000							
65		Add: Cash collections	170,000							
66		Total cash available	210,000							
67		Less: Cash disbursements								
68		Materials	40,000							
69		Direct labor	15,000							
70		Manufacturing overhead	56,000							
71		Selling and administrative	70,000							
72		Equipment purchase	-							
73		Dividend	49,000							
74		Total disbursements	230,000							
75		Excess (deficiency)	(20,000)							
76		Financing:								
77		Borrowing	50,000							
78		Repayments	-							
79		Interest	-							
80		Total financing	50,000							
81		Ending cash balance	\$ 30,000							
82										

Ending cash balance for April is the beginning May balance.

The Cash Budget

M88									
	A	B	C	D	E	F	G	H	I
63			April		May		June		Quarter
64		Beginning cash balance	\$ 40,000		\$ 30,000				
65		Add: Cash collections	170,000		400,000				
66		Total cash available	210,000		430,000				
67		Less: Cash disbursements							
68		Materials	40,000		72,300				
69		Direct labor	15,000		23,000				
70		Manufacturing overhead	56,000		76,000				
71		Selling and administrative	70,000		85,000				
72		Equipment purchase	-		143,700				
73		Dividend	49,000		-				
74		Total disbursements	230,000		400,000				
75		Excess (deficiency)	(20,000)		30,000				
76		Financing:							
77		Borrowing	50,000		-				
78		Repayments	-		-				
79		Interest	-		-				
80		Total financing	50,000		-				
81		Ending cash balance	\$ 30,000		\$ 30,000				
82									

Quick Check ✓

What is the excess (deficiency) of cash available over disbursements for June?

- a. \$ 85,000
- b. \$(10,000)
- c. \$ 75,000
- d. \$ 95,000



Quick Check ✓

What is the excess (deficiency) of cash available over disbursements for June?

- a. \$ 85,000
- b. \$(10,000)
- c. \$ 75,000
- d. \$ 95,000**



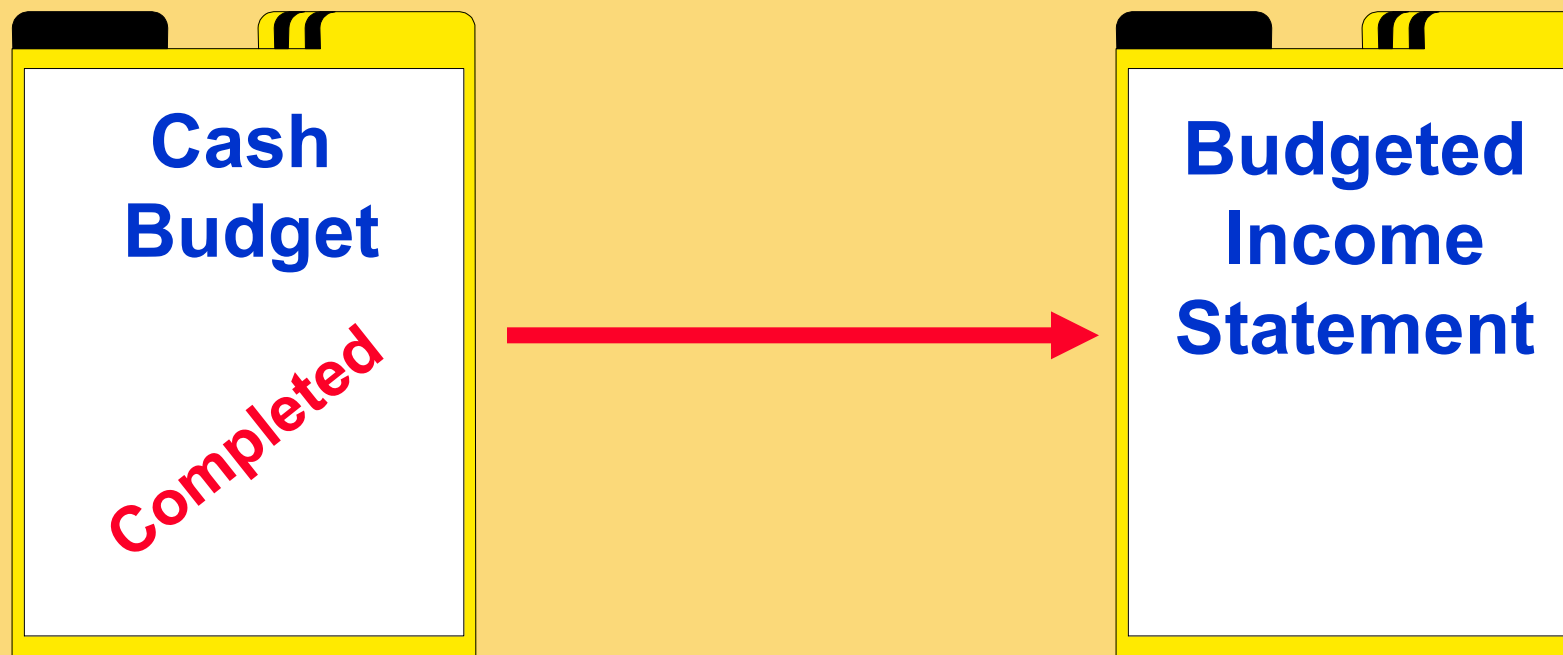
The Cash Budget

M88									
	A	B	C	D	E	F	G	H	I
63			April		May		June		Quarter
64	Beginning cash balance		\$ 40,000		\$ 30,000		\$ 30,000		\$ 40,000
65	Add: Cash collections		170,000		400,000		335,000		905,000
66	Total cash available		210,000		430,000		365,000		945,000
67	Less: Cash disbursements								
73					72,300		72,700		185,000
74					23,000		15,000		53,000
75					76,000		59,000		191,000
76					85,000		75,000		230,000
77					143,700		48,300		192,000
78					-		-		49,000
79	Total disbursements		230,000		400,000		270,000		900,000
80	Excess (deficiency)		(20,000)		30,000		95,000		45,000
81	Financing:								
82	Borrowing		50,000		-		-		50,000
83	Repayments		-		-		(50,000)		(50,000)
84	Interest		-		-		(2,000)		(2,000)
85	Total financing		50,000		-		(52,000)		(2,000)
86	Ending cash balance		\$ 30,000		\$ 30,000		\$ 43,000		\$ 43,000

$$\text{\$50,000} \times 16\% \times 3/12 = \text{\$2,000}$$

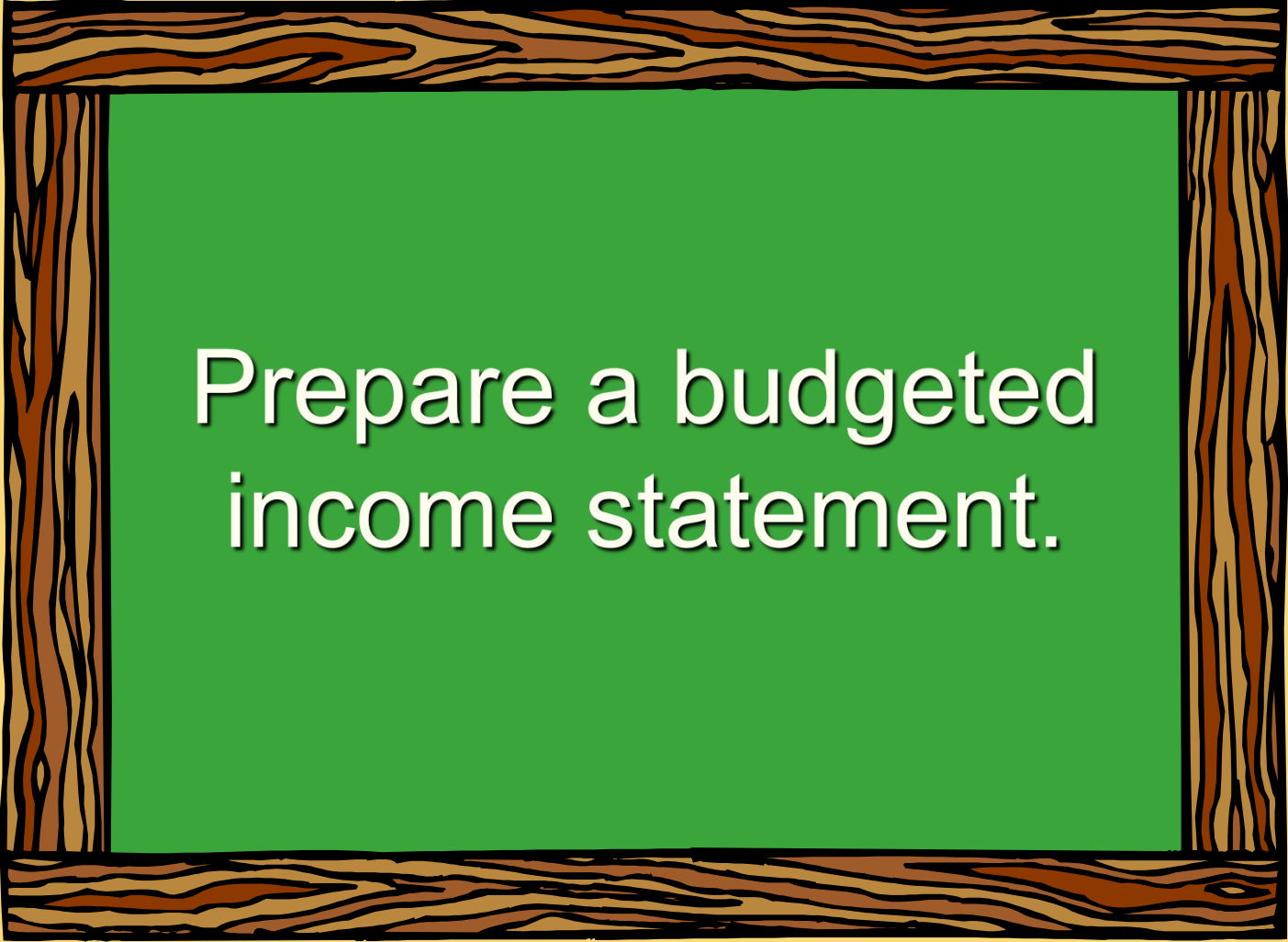
Borrowings on April 1 and
repayment on June 30.

The Budgeted Income Statement



**After we complete the cash budget,
we can prepare the budgeted income
statement for Royal.**

Learning Objective 9



Prepare a budgeted
income statement.

The Budgeted Income Statement

Royal Company Budgeted Income Statement For the Three Months Ended June 30

Sales (100,000 units @ \$10)	\$ 1,000,000
Cost of goods sold (100,000 @ \$4.99)	499,000
Gross margin	501,000
Selling and administrative expenses	260,000
Operating income	241,000
Interest expense	2,000
Net income	\$ 239,000

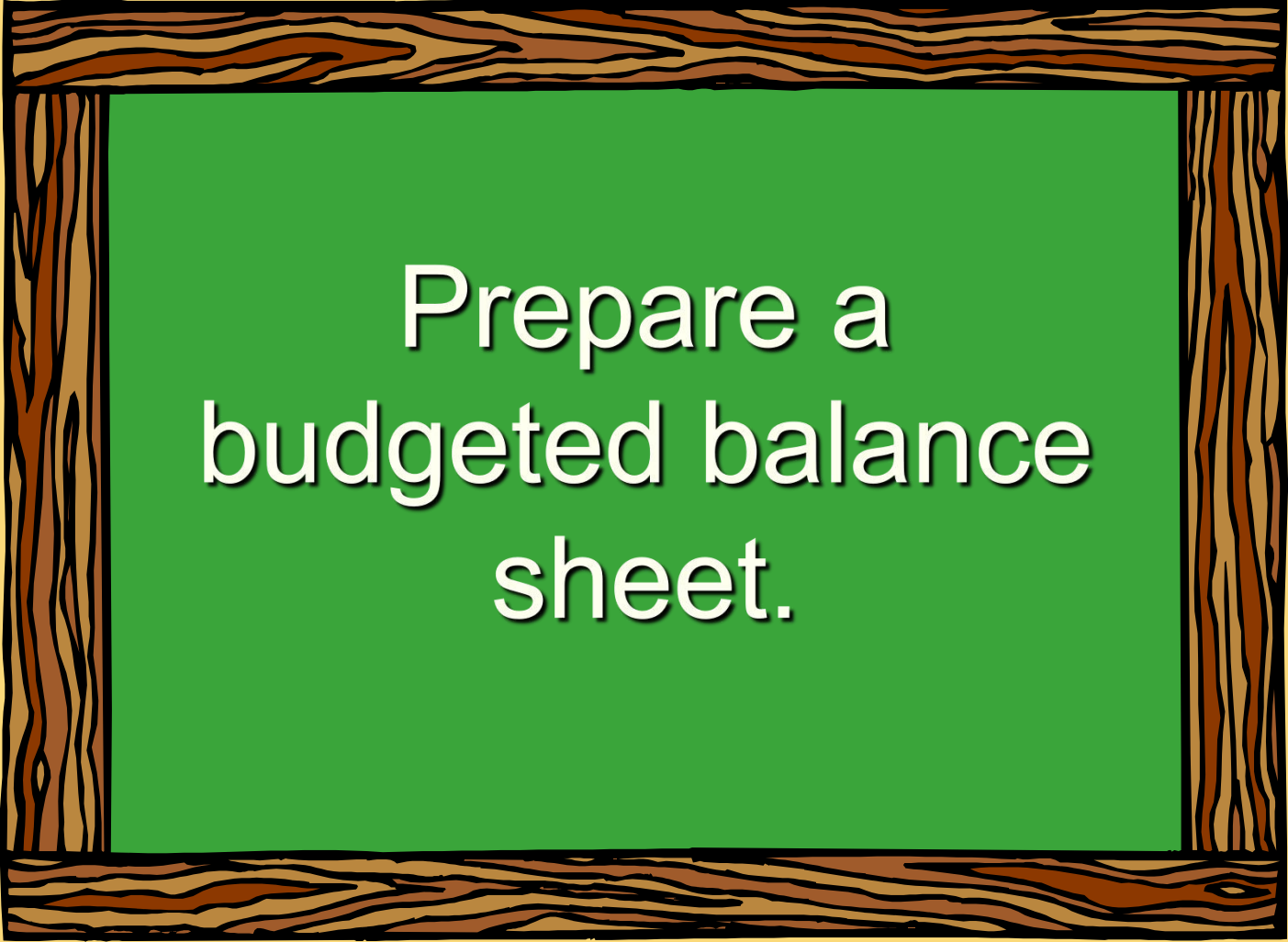
Sales Budget.

Ending Finished
Goods Inventory.

Selling and
Administrative
Expense Budget.

Cash Budget.

Learning Objective 10



Prepare a
budgeted balance
sheet.

The Budgeted Balance Sheet

Royal reported the following account balances prior to preparing its budgeted financial statements:

- Land - \$50,000
- Common stock - \$200,000
- Retained earnings - \$146,150
- Equipment - \$175,000

Royal Company
Budgeted Balance Sheet
June 30

Current assets

Cash \$ 43,000

Accounts receivable 75,000

Raw materials inventory 4,600

Finished goods inventory 24,950

Total current assets 147,550

Property and equipment

Land 50,000

Equipment 367,000

Total property and equipment 417,000

Total assets \$ 564,550

Accounts payable \$ 28,400

Common stock 200,000

Retained earnings 336,150

Total liabilities and equities \$ 564,550

25% of June
sales of
\$300,000.

11,500 lbs.
at \$0.40/lb.

5,000 units
at \$4.99 each.

50% of June
purchases
of \$56,800.

Royal Company

Budgeted Balance Sheet

June 30

Current assets

Cash

\$

Accounts receivable

Raw materials inventory

Finished goods inventory

Total current assets

Property and equipment

Land

50,000

Equipment

367,000

Total property and equipment

417,000

Total assets

\$ 564,550

Accounts payable

\$ 28,400

Common stock

200,000

Retained earnings

336,150

Total liabilities and equities

\$ 564,550

Beginning balance	\$146,150
Add: net income	239,000
Deduct: dividends	<u>(49,000)</u>
Ending balance	<u><u>\$336,150</u></u>

End of Chapter 9

